Structural Design Report

1. Executive Summary

Total Floor Area	108.00 m²
Stories / Height	2 Stories / 6.0 m
RCC Concrete Volume	66.70 m³
Reinforcement Steel	4320 kg
Clay Modular Bricks	17,602 Nos
Masonry & Plaster Cement	158 Bags (50kg)
Masonry & Plaster Sand	38.48 Tonnes
Plaster Surface Area	703.1 m²
Total Estimated Cost	INR 1,650,378.46

2. Design Basis

Design Codes & Standards:
- IS 456:2000 (Code of Practice for Plain and Reinforced Concrete)

Material Properties:
- Concrete: M25 (Density: 25 kN/m3, E = 5000sqrt(fck))
- Steel: Fe415 (Density: 78.5 kN/m3, E = 2x10^5 MPa)

Loading & Analysis:
- Design Load: User Input (Factored)
- Load Path: Tributary Area Method (Corner: 25%, Edge: 50%, Interior: 100% of bay)
- Beam Analysis: Simplified Simply Supported/Continuous assumptions.
(M = wL^2/8 or wL^2/10, Deflection checked against L/250)

Column Design:
- Short Column Assumption (Le/r < 12)
- Sizing Criteria: Axial Stress Check (Pu/Ag <= 0.4 fck)
Note: Steel capacity (0.67 fy Asc) is reserved as safety margin.

Foundation Design:
- Isolated Square Footings
- Area Required = Load * 1.1 / SBC (1.1 factor for self-weight)
- Depth Check: Punching Shear at d/2 distance (Limit: 0.25sqrt(fck))

Estimation Basis:
- Concrete Cost: INR 5000/m3
- Steel Cost: INR 60/kg
- Rebar Ratios: Columns 150kg/m3, Beams 120kg/m3, Footings 80kg/m3

3. Column Schedule & Quantities (IS 456 Cl. 26.5.3)

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C1_L0	L0	(0.00, 0.00)	185.2	300x450	0.41	60.8	PASS

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C1_L1	L1	(0.00, 0.00)	92.6	300x450	0.41	60.8	PASS
C2_L0	L0	(0.00, 4.50)	311.2	300x450	0.41	60.8	PASS
C2_L1	L1	(0.00, 4.50)	155.6	300x450	0.41	60.8	PASS
C3_L0	L0	(0.00, 9.00)	185.2	300x450	0.41	60.8	PASS
C3_L1	L1	(0.00, 9.00)	92.6	300x450	0.41	60.8	PASS
C4_L0	L0	(4.00, 0.00)	305.2	300x450	0.41	60.8	PASS
C4_L1	L1	(4.00, 0.00)	152.6	300x450	0.41	60.8	PASS
C5_L0	L0	(4.00, 4.50)	505.5	400x600	0.72	108.0	PASS
C5_L1	L1	(4.00, 4.50)	252.8	400x600	0.72	108.0	PASS
C6_L0	L0	(4.00, 9.00)	305.2	300x450	0.41	60.8	PASS
C6_L1	L1	(4.00, 9.00)	152.6	300x450	0.41	60.8	PASS
C7_L0	L0	(8.00, 0.00)	305.2	300x450	0.41	60.8	PASS
C7_L1	L1	(8.00, 0.00)	152.6	300x450	0.41	60.8	PASS
C8_L0	L0	(8.00, 4.50)	505.5	400x600	0.72	108.0	PASS
C8_L1	L1	(8.00, 4.50)	252.8	400x600	0.72	108.0	PASS
C9_L0	L0	(8.00, 9.00)	305.2	300x450	0.41	60.8	PASS
C9_L1	L1	(8.00, 9.00)	152.6	300x450	0.41	60.8	PASS
C10_L0	L0	(12.00, 0.00)	185.2	300x450	0.41	60.8	PASS
C10_L1	L1	(12.00, 0.00)	92.6	300x450	0.41	60.8	PASS
C11_L0	L0	(12.00, 4.50)	311.2	300x450	0.41	60.8	PASS
C11_L1	L1	(12.00, 4.50)	155.6	300x450	0.41	60.8	PASS
C12_L0	L0	(12.00, 9.00)	185.2	300x450	0.41	60.8	PASS
C12_L1	L1	(12.00, 9.00)	92.6	300x450	0.41	60.8	PASS

4. Reinforcement Schedule (IS 456 Cl. 26.5)

Col ID	Main Bars	Links (Stirrups)
C1_L0	8-16#	8# @ 150 c/c
C1_L1	8-16#	8# @ 200 c/c
C2_L0	8-16#	8# @ 150 c/c
C2_L1	8-16#	8# @ 200 c/c
C3_L0	8-16#	8# @ 150 c/c
C3_L1	8-16#	8# @ 200 c/c
C4_L0	8-16#	8# @ 150 c/c
C4_L1	8-16#	8# @ 200 c/c
C5_L0	10-16#	8# @ 150 c/c
C5_L1	10-16#	8# @ 200 c/c

Col ID	Main Bars	Links (Stirrups)
C6_L0	8-16#	8# @ 150 c/c
C6_L1	8-16#	8# @ 200 c/c
C7_L0	8-16#	8# @ 150 c/c
C7_L1	8-16#	8# @ 200 c/c
C8_L0	10-16#	8# @ 150 c/c
C8_L1	10-16#	8# @ 200 c/c
C9_L0	8-16#	8# @ 150 c/c
C9_L1	8-16#	8# @ 200 c/c
C10_L0	8-16#	8# @ 150 c/c
C10_L1	8-16#	8# @ 200 c/c
C11_L0	8-16#	8# @ 150 c/c
C11_L1	8-16#	8# @ 200 c/c
C12_L0	8-16#	8# @ 150 c/c
C12_L1	8-16#	8# @ 200 c/c

5. Foundation Schedule & Quantities (IS 456 Cl. 34)

Col ID	Load (kN)	Size (m)	Depth (mm)	Conc (m3)	Steel (kg)
C1_L0	185.2	1.50 x 1.50	300	0.67	54.0
C2_L0	311.2	1.50 x 1.50	300	0.67	54.0
C3_L0	185.2	1.50 x 1.50	300	0.67	54.0
C4_L0	305.2	1.50 x 1.50	300	0.67	54.0
C5_L0	505.5	2.00 x 2.00	350	1.40	112.0
C6_L0	305.2	1.50 x 1.50	300	0.67	54.0
C7_L0	305.2	1.50 x 1.50	300	0.67	54.0
C8_L0	505.5	2.00 x 2.00	350	1.40	112.0
C9_L0	305.2	1.50 x 1.50	300	0.67	54.0
C10_L0	185.2	1.50 x 1.50	300	0.67	54.0
C11_L0	311.2	1.50 x 1.50	300	0.67	54.0
C12_L0	185.2	1.50 x 1.50	300	0.67	54.0

6. Beam Schedule & Quantities (IS 456 Cl. 26.5.1)

ID	Size (mm)	Span (m)	Conc (m3)	Steel (kg)	Top Steel	Bot Steel	Stirrups
B_H_1	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_2	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_3	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_4	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c

ID	Size (mm)	Span (m)	Conc (m3)	Steel (kg)	Top Steel	Bot Steel	Stirrups
B_H_5	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_6	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_7	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_8	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_9	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_V_10	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_11	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_12	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_13	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_14	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_15	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_16	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_17	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c

7. Slab Schedule & Quantities

ID	Thk (mm)	Area (m2)	Conc (m3)	Steel (kg)	Main Mesh	Dist Mesh
S1	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S2	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S3	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S4	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S5	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S6	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c

8. Masonry Wall Design & Material Takeoff

8.1 Individual Wall Section Material Takeoff

Quantities per wall segment per floor: Bricks = 250 × Vol (IS 1077), Mortar dry factor = 1.33 (IS 2250), Plaster 12/15mm (IS 1661/2402).

Wall ID	Span / Cols	Len (m)	Ht (m)	Thk	Vol (m3)	Bricks	Mortar (m3)	Cem (kg)	Sand (T)	Plaster (m2)
W_H1	C1_L0→C4_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H2	C4_L0→C7_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H3	C7_L0→C10_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H4	C2_L0→C5_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H5	C5_L0→C8_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H6	C8_L0→C11_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H7	C3_L0→C6_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_H8	C6_L0→C9_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8

Wall ID	Span / Cols	Len (m)	Ht (m)	Thk	Vol (m3)	Bricks	Mortar (m3)	Cem (kg)	Sand (T)	Plaster (m2)
W_H9	C9_L0→C12_L0	4.00	2.48	200mm	1.98	495	0.372	142.4	0.63	19.8
W_V10	C1_L0→C2_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V11	C2_L0→C3_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V12	C4_L0→C5_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V13	C5_L0→C6_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V14	C7_L0→C8_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V15	C8_L0→C9_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V16	C10_L0→C11_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3
W_V17	C11_L0→C12_L0	4.50	2.48	200mm	2.23	557	0.418	160.0	0.71	22.3

8.2 Overall Masonry & Finishes Material Summary (All Stories)

Item / Material	Specification / Mix	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear height)	71.28	m³	IS 1905:1987
Clay Building Bricks	Modular 20×20×10 cm (with mortar)	17,822	Nos	IS 1077:1992
Wet Mortar Volume	Wall Vol – Solid Brick Vol	13.38	m³	IS 2250:1981
Dry Mortar Volume	Wet Volume × 1.33 factor	17.79	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion	103 (5,124 kg)	Bags (50kg)	IS 269:2015
Brickwork Sand	Dry bulk density 1600 kg/m³	22.77	Tonnes	IS 383 Zone II
Masonry Water	W:C ratio = 0.50	2,562	Litres	IS 456 Cl 5.4
Total Plaster Surface Area	Both sides (12mm int / 15mm ext)	712.8	m²	IS 1200 Pt 12
Plastering Cement	1:6 internal, 1:4 external	58 (2,852 kg)	Bags (50kg)	IS 1661 / IS 2402
Plastering Sand	Clean river / M-sand	16.36	Tonnes	IS 383 Zone III
Plastering Water	W:C ratio = 0.50	1,426	Litres	IS 456 Cl 5.4
Surface Primer	1 coat (25–40 microns) @ 8 m²/L	89.1	Litres	NBC 2016
Wall Putty	2 coats (1–2 mm) @ 1.5 kg/m²	1,514.7	kg	NBC 2016
Interior Acrylic Emulsion	2 coats (30–40 microns) @ 12 m²/L	84.1	Litres	NBC 2016
Exterior Weather Emulsion	2 coats (40–60 microns) @ 10 m²/L	41.6	Litres	NBC 2016
TOTAL CEMENT (Masonry + Plaster)	Mortar + Plaster combined	161 bags (8,050 kg)	Bags	IS 269
TOTAL SAND (Masonry + Plaster)	Mortar + Plaster combined	39.13	Tonnes	IS 383

9. Structural Layout Plan

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Column Coordinates: C1_L0: (0.00, 0.00) C2_L0: (0.00, 4.50) C3_L0: (0.00, 9.00)
C4_L0: (4.00, 0.00) C5_L0: (4.00, 4.50) C6_L0: (4.00, 9.00) C7_L0: (8.00, 0.00)
C8_L0: (8.00, 4.50) C9_L0: (8.00, 9.00) C10_L0: (12.00, 0.00) C11_L0: (12.00, 4.50)
C12_L0: (12.00, 9.00) C1_L1: (0.00, 0.00) C2_L1: (0.00, 4.50) C3_L1: (0.00, 9.00)
C4_L1: (4.00, 0.00) C5_L1: (4.00, 4.50) C6_L1: (4.00, 9.00) C7_L1: (8.00, 0.00)
C8_L1: (8.00, 4.50) C9_L1: (8.00, 9.00) C10_L1: (12.00, 0.00) C11_L1: (12.00, 4.50)
C12_L1: (12.00, 9.00)
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10. Integrated Steel Breakdown

Bar Diameter	Total Weight (kg)
8 mm	561.3
10 mm	1717.4
12 mm	132.8
16 mm	984.9
20 mm	923.3
TOTAL	4319.8

10. Structural Audit Report

No audit results available.

11. Sustainability Audit

Design Strategy: Standard (OPC)

Material	Quantity	Emission Factor	Total CO2e (kg)
Concrete	66.70 m3	300 kg/m3	20010.0
Steel	4320 kg	1.85 kg/kg	7991.6
TOTAL	-	-	28001.6

Wait! You can reduce your Carbon Footprint.
Refining the design to use **Fly Ash based Concrete (PPC)** instead of OPC can reduce concrete emissions by approximately 15-20%.
Switch "Use Green Concrete" ON in the dashboard to see savings. (Note: PPC is slower setting and has lower early strength, use only if confirmed by RMC supplier).

12. Serviceability Checks (Deflection)

No beam serviceability checks performed.

13. Staircase Design Schedule (Dog-Legged)

No staircase selected for design.

Concrete Material Breakdown (IS 10262:2019)

Material	Per m³	Total
Wet Concrete Volume		66.70 m³
Cement (50kg bags)	7.6 bags	506.9 bags
Fine Aggregate (Sand)	0.419 m³	27.93 m³
Coarse Aggregate (20mm)	0.821 m³	54.74 m³
Water	171 kg	11406 kg
Water:Cement Ratio	0.45	—
Nominal Mix Ratio	1:1.76:3.13	—

Note: Indicative quantities per IS 10262:2019. Add 3-5% wastage for actual procurement.

Beam L/d Deflection Check (IS 456 Cl 23.2)

Beam ID	Span (mm)	Depth (mm)	Actual L/d	Allowable	Status
B_H_1	4000	350	11.4	20	PASS
B_H_2	4000	350	11.4	20	PASS
B_H_3	4000	350	11.4	20	PASS
B_H_4	4000	350	11.4	20	PASS
B_H_5	4000	350	11.4	20	PASS
B_H_6	4000	350	11.4	20	PASS
B_H_7	4000	350	11.4	20	PASS
B_H_8	4000	350	11.4	20	PASS
B_H_9	4000	350	11.4	20	PASS
B_V_10	4500	400	11.2	20	PASS
B_V_11	4500	400	11.2	20	PASS
B_V_12	4500	400	11.2	20	PASS
B_V_13	4500	400	11.2	20	PASS
B_V_14	4500	400	11.2	20	PASS
B_V_15	4500	400	11.2	20	PASS
B_V_16	4500	400	11.2	20	PASS
B_V_17	4500	400	11.2	20	PASS

14. Professional Disclaimer

IMPORTANT NOTICE - READ CAREFULLY

This structural design report is generated by automated software for **PRELIMINARY DESIGN AND EDUCATIONAL PURPOSES ONLY**.

Key Limitations:
• The software uses simplified analysis methods (tributary area, short column assumptions).
• Seismic forces (IS 1893) and Wind loads (IS 875 Pt 3) have been incorporated for basic checks, but require full dynamic analysis for high-rise buildings.
• Site-specific soil conditions, groundwater, and other geotechnical factors are not evaluated.
• Local building codes may have additional requirements beyond IS 456:2000.

Required Actions Before Construction:
• All designs MUST be reviewed and sealed by a licensed Professional Engineer (PE).
• Independent structural analysis should be performed using appropriate software.
• Geotechnical investigation and foundation recommendations are required.
• All applicable permits and approvals must be obtained.

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All designs must be verified by a licensed Structural Engineer prior to construction.
Safety Factors: γc = 1.5 (concrete), γs = 1.15 (steel) | Load Factor: 1.5 (DL + LL)
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